TECHNOLOGY ETHICS • COMPLETE STUDY NOTES

Set 7: Privacy

Privacy vs Security Balance, Government Surveillance, Smart Lamppost Case, Data Mining, Facial Recognition & Workplace Monitoring

PART 7

Privacy

3 sections, about 1 h 45 min of reading.

Privacy, the law and government surveillance

~40 min to read | Medium priority | 9 questions

Source: Set 7 (Lecture 4: Privacy), slides 1-8 and the legal summary on slide 28. Continues in encryption and identity theft and profiling, workplace monitoring and advanced surveillance.

Privacy protection and the law: the ethical conundrum

The lecture opens with a **conundrum**. A conundrum is a puzzling question: a problem with no easy answer because two good things pull against each other. Here the two things are the needs of those who use information and the privacy rights of the people the information is about.

The slides set the problem out in four steps.

1. **IT technology allows businesses to gather information.** Gathering is cheap, fast and automatic.
2. **The users of this information must be balanced against the privacy rights of the people whose information may be used.** That is the conundrum itself: the organisation has a legitimate need, the person has a legitimate right.
3. **Systems collect and store key data from every interaction with customers.** Examples on the slide: purchasing habits, contacts, search terms. Every click, purchase and search leaves a record.
4. **Many people object to the data-collection policies of government and business.** The objection is that collection strips people of the power to control their own personal information. The slide adds a dry remark: IT does this on a regular basis. In other words, the objection is serious, but the practice is routine.

> **Key point.** The core ethical complaint about data collection is loss of *control*. The person no longer decides who holds their information or what it is used for. Use the words "power to control their own personal information" in an essay.

Why privacy matters, reasonable limits and the Fourth Amendment

Slide 3 continues in three moves.

- **Privacy is a key concern of Internet users.** It is also the top reason why non-users still avoid the Internet, according to US Census data. So weak privacy protection does not only hurt users. It also keeps people away from the technology and its benefits.
- **Reasonable limits must be set.** Information and communication technologies must be developed to *protect* privacy rather than diminish it. This is the lecturer's conclusion on the conundrum: neither side wins outright, but limits are needed, and technology should be designed with privacy in mind.
- **Historical perspective on the right to privacy.** The slide names the **Fourth Amendment** (dated 1789 on the slide) of the US Constitution. It gives a *reasonable expectation of privacy* and protects against *unreasonable searches and seizures*. beyond the slides The Bill of Rights was in fact ratified in 1791. Use the slide's wording for the exam, but do not be surprised if a textbook gives a different date.

The two vocabulary items from slide 7 belong here. The **Fourth Amendment** is the right to be protected from unreasonable search. The **First Amendment** is freedom of speech, and the like. The First Amendment matters in privacy debates because anonymity and private communication help people speak freely, and surveillance can make them hold back (a "chilling" effect).

> **Watch out.** The Fourth Amendment is a US law. It restrains the government, not private employers (see workplace monitoring in the next section). In a Ugandan exam, present it as the historical source of the idea of privacy, then argue the principle, not the US legal detail.

Q: Slides 2-3 · Practice · 3 marks **What is the ethical conundrum in business and government collection of personal data?**

MARKER WORDS: THE CONUNDRUM AND THE RIGHT TO PRIVACY

Say it like this:

> IT lets organisations **gather information**, so they must **balance** their needs against the **privacy rights** of the people concerned, because collection **strips people of the power to control** their own personal information.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
right to privacy	naming what is at stake in any data or monitoring question	**MUST**
balance	framing any privacy-versus-security or privacy-versus-business answer	**MUST**
reasonable limits	giving your conclusion	**SHOULD**
reasonable expectation of privacy	explaining the Fourth Amendment idea	**SHOULD**
unreasonable searches and seizures	stating what the Fourth Amendment protects against	**SHOULD**
control their own personal information	explaining why people object to collection	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Conundrum
A puzzling question.

Fourth Amendment
The right to be protected from unreasonable search, etc. (reasonable expectation of privacy; protection against unreasonable searches and seizures).

First Amendment
Freedom of speech, etc.

What the ethical conundrum of IT-based data gathering balances.
The needs of those who use the information against the privacy rights of the people whose information may be used.

Key privacy and anonymity issues

Slide 4 is a list of eight issues. This lecture treats each one in turn, so the list is also a map of the whole privacy topic. Learn it whole.

1. Government electronic surveillance (this section)
2. Data encryption (next section)
3. Identity theft (next section)
4. Customer profiling (after that)
5. Need to treat customer data responsibly (after that)
6. Workplace monitoring (after that)
7. Spamming (after that)
8. Advanced surveillance techniques (after that)

The slide title joins privacy with **anonymity**. Privacy is control over personal information. Anonymity is being able to act, browse or speak without being identified. Encryption and anonymous browsing protect it. Surveillance and profiling erode it.

FLASHCARDS: CLICK TO FLIP

The eight key privacy and anonymity issues.
Government electronic surveillance; data encryption; identity theft; customer profiling; need to treat customer data responsibly; workplace monitoring; spamming; advanced surveillance techniques.

Governmental electronic surveillance and the USA Patriot Act of 2001

Slide 5 is the main legal slide. Its facts, in order:

- **Full title.** The USA PATRIOT Act of 2001 is "Uniting and Strengthening America by Providing Appropriate Tools Required to Intercept and Obstruct Terrorism". The word PATRIOT is an acronym built from these words. Learn the full title: it can be asked.
- **Passed just after 9/11.** It is more than 340 pages long and quite complex, yet it passed into law only **five weeks after being introduced**. The point to draw out: very little time for debate on a very large law.
- **Sweeping new powers** to two groups: *domestic law enforcement*, including **wiretaps without court order**, and *international intelligence agencies*.
- **Critics' argument.** The Act removes the **checks and balances** that previously gave courts the opportunity to ensure that law enforcement agencies did not abuse their powers. A court order was the check. Take it away and nobody independent confirms that a wiretap is justified.

- **Sunset provisions.** The Act contains several "sunset" provisions covering increased searches and electronic surveillance. A sunset provision is a clause that makes a power expire on a set date unless it is renewed. These ones were due to terminate on 12/31/05 (31 December 2005).
- **An amendment to FISA.** The Act amends the **Foreign Intelligence Surveillance Act** (FISA, 1978; see the legal summary below).

Provisions subject to sunset (slides 6 and 8) and the terms used (slide 7)

Slides 6 and 8 are two tables, headed "Key Provisions of USA Patriot Act Subject to Sunset" (slide 6) and "... (continued)" (slide 8). They list the sections of the Act that were due to expire. The slide 6 caption says they are **primarily those that modified Title III and FISA**. Together the two slides give this list.

Section	Issue addressed	Summary
201	Wiretapping in terrorism cases	Added several crimes for which federal courts may authorize wiretapping of people's communications.
202	Wiretapping in computer fraud and abuse felony cases	Added computer fraud and abuse to the list of crimes the FBI may obtain a court order to investigate under Title III.
203 b	Sharing wiretap information	Allows the FBI to disclose evidence obtained under Title III to other federal officials, including "law enforcement, intelligence, protective, immigration, national defense, [and] national security" officials.
203 d	Sharing foreign intelligence information	Provides for disclosure of threat information obtained during criminal investigations to "appropriate" federal, state, local, or foreign government officials for the purpose of responding to the threat.
204	FISA pen register/trap-and-trace exceptions	Exempts foreign intelligence surveillance from statutory prohibitions against the use of pen register or trap-and-trace devices, which capture "addressing" information about the sender and recipient of a communication. It also exempts the U.S. government from general prohibitions against intercepting electronic communications and allows stored voice-mail communication to be obtained by the government through a search warrant rather than more stringent wiretap orders.
206	FISA roving wiretaps	Expands FISA to permit "roving wiretap" authority, which allows the FBI to intercept any communications to or by an intelligence target without specifying the telephone line, computer, or other facility to be monitored.
207	Duration of FISA surveillance of non-U.S. agents of a foreign power	Extends the duration of FISA wiretap orders relating to an agent of a foreign power from 90 days to 120 days, and allows an extension in 1-year intervals instead of 90-day increments.
209	Seizure of voice-mail messages pursuant to warrants	Enables the government to obtain voice-mail messages under Title III using just a search warrant rather than a wiretap order, which is more difficult to obtain. Messages stored on an answering machine tape, however, remain outside the scope of this section.
212	Emergency disclosure of electronic surveillance	Permits providers of communication services (such as telephone companies and Internet service providers) to disclose consumer records to the FBI if they believe immediate danger of serious physical injury is involved. Communication providers cannot be sued for such disclosure.

Section	Issue addressed	Summary
214	FISA pen register/trap-and-trace authority	Allows the government to obtain a pen register/trap-and-trace device "for any investigation to gather foreign intelligence information." It prohibits the use of FISA pen register/trap-and-trace surveillance against a U.S. citizen when the investigation is conducted "solely on the basis of activities protected by the First Amendment."
215	FISA access to tangible items	Permits the FBI to compel production of any record or item without showing probable cause. People served with a search warrant issued under FISA rules may not disclose, under penalty of law, the existence of the warrant or the fact that records were provided to the government. It prohibits investigation of a U.S. citizen when it is conducted solely on the basis of activities protected by the First Amendment.
217	Interception of computer trespasser communications	Creates a new exception to Title III that permits the government to intercept the "communications of a computer trespasser" if the owner or operator of a "protected computer" authorizes it. It defines a protected computer as any computer "used in interstate or foreign commerce or communication" (because of the Internet, this effectively includes almost every computer).
220	Nationwide service of search warrants for electronic evidence	Expands the geographic scope where the FBI can obtain search warrants or court orders for electronic communications and customer records.
223	Civil liability and discipline for privacy violations	Provides that people can sue the government for unauthorized disclosure of information obtained through surveillance.
225	Provider immunity for FISA wiretap assistance	Provides immunity from lawsuits for people who disclose information to the government pursuant to a FISA wiretap order, physical search order, or an emergency wiretap or search.

> **Pattern to remember.** More crimes covered, more sharing between agencies, longer orders, and less need for a court order or probable cause. The only privacy-side clauses are 223 (the right to sue) and the First Amendment limits in 214 and 215.

The caption on slide 6 says the sunset provisions are **primarily those that modified Title III and FISA**. The slide explains what that means: these changes **allow the government to conduct wiretaps without a court order**.

- **Title III** is the part of an older US law that sets the rules for police wiretapping of ordinary crime suspects. `beyond the slides` It is the Omnibus Crime Control and Safe Streets Act of 1968. It normally

requires a judge's permission before a phone is tapped.

- **FISA** is the law that governs surveillance for foreign intelligence purposes (wiretapping aliens, per slide 28).
- The Patriot Act **modified** both so that more surveillance could happen with *less* court involvement. These modified parts were the ones with the expiry date.

Slide 7 gives the terms needed to read this material.

Term	Meaning on the slide
Pen register	Collects the *outgoing* phone numbers placed from a specific telephone line.
Trap and trace device	Captures the *incoming* numbers placed to a specific phone line. Example: a caller-ID box is a trap and trace device.
First Amendment	Freedom of speech, etc.
Fourth Amendment	Right to be protected from unreasonable search, etc.
Conundrum	A puzzling question.

Together, a pen register and a trap and trace device record *who called whom* (not what was said). This is called "metadata" today. **beyond the slides** Even without content, a call log reveals a lot about a person's life.

Weighing the Patriot Act: the two sides

The slides state the critics' side and the purpose of the Act. Put them together and you have a ready-made two-sided argument, useful for any "privacy versus security" essay.

For stronger surveillance powers	Against (critics)
Aim is to intercept and obstruct terrorism, after the 9/11 attacks.	Removes checks and balances that let courts stop abuse of power.
Agencies need new tools to track modern communications.	Wiretaps without a court order, with no independent review.
Sunset clauses make the extra powers temporary and force a review.	Passed in five weeks, 340+ pages, little time to debate.
Public safety is a basic duty of government.	Threatens the Fourth Amendment's protection against unreasonable search.

MARKER WORDS: THE PATRIOT ACT AND ELECTRONIC SURVEILLANCE

Say it like this:

> The USA Patriot Act of 2001, passed **five weeks** after 9/11, gave law enforcement and intelligence agencies **sweeping new powers**, including **wiretaps without court order**. Critics say it removes the **checks and balances** that let courts stop **abuse**. Its surveillance provisions carried **sunset** clauses and amend **FISA**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
government electronic surveillance	naming the issue	**SHOULD**
sweeping new powers	describing the Patriot Act	**SHOULD**
wiretaps without court order	giving the main power and main criticism	**SHOULD**
checks and balances	explaining what critics say was removed	**SHOULD**
abuse their powers	explaining the risk of surveillance with no review	**SHOULD**
sunset provisions	explaining that extra powers were meant to be temporary	**SHOULD**
pen register / trap and trace	defining the call-recording devices	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Full title of the USA Patriot Act of 2001.
Uniting and Strengthening America by Providing Appropriate Tools Required to Intercept and Obstruct Terrorism.

Pen register
Collects the outgoing phone numbers placed from a specific telephone line.

Trap and trace device
Captures the incoming numbers placed to a specific phone line (a caller-ID box is an example).

Sunset provisions in the Patriot Act
Provisions for increased searches and electronic surveillance that terminated on 12/31/05; primarily those that modified Title III and FISA (allow government to conduct wiretaps without court order).

Four facts about the Patriot Act's passage and powers.
Passed just after 9/11; over 340 pages but passed five weeks after introduction; sweeping new powers to domestic law enforcement (wiretaps without court order) and international intelligence agencies; amendment to FISA.

Legal summary: the right to privacy and the main surveillance laws (slide 28)

Slide 28 is the lecture summary. Two parts of it belong to this section.

1. The legal concept of the right to privacy has four aspects. The slide lists them in shorthand:

1. **Web-surfing kept private**: what you browse should not be tracked and exposed without your say.
2. **Identity theft**: protection against others using your personal data in your name.
3. **HIPPA** (spelled so on the slide; the US health-records privacy law is normally written HIPAA): privacy of medical information.
4. **False information about someone on the web**: protection against untrue information being published about a person.

2. A number of laws enacted over the past 40 years affect a person's privacy. The three named on the slide:

Law	Year	What the slide says
FISA (Foreign Intelligence Surveillance Act)	1978	Wiretapping aliens.
ECPA (Electronic Communications Privacy Act)	1986	Set standards for access to stored email and other electronic communication.
US PATRIOT Act	2001	Authorises surveillance powers (see above).

The slide's conclusion: these and other laws **authorize electronic surveillance by the government**. So the law both protects privacy (ECPA sets standards for access to stored communication) and permits intrusion (FISA, Patriot Act). That double role is why the right to privacy is always argued as a balance. The rest of slide 28 (encryption, identity theft) is in the next section, and the profiling, workplace and surveillance summary on slide 29 is in the section after.

MARKER WORDS: PRIVACY LAW SUMMARY

Say it like this:

> Laws such as **FISA 1978** (wiretapping aliens), **ECPA 1986** (standards for access to stored email and electronic communication) and the **USA Patriot Act 2001** **authorize electronic surveillance by the government**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
right to privacy	opening any privacy answer	**MUST**
four aspects	listing what the legal right covers	**SHOULD**
FISA 1978	naming surveillance law	**SHOULD**
ECPA 1986	naming the stored-communication law	**SHOULD**
authorize electronic surveillance	stating the law's overall effect	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The four aspects of the legal right to privacy (slide 28).
Web-surfing private; ID theft; HIPPA (health information); false info about someone on the web.

FISA 1978
Foreign Intelligence Surveillance Act: wiretapping aliens.

ECPA 1986
Electronic Communications Privacy Act: set standards for access to stored email and other electronic communication.

Three privacy laws on the summary slide.
FISA 1978 (wiretapping aliens); ECPA 1986 (access to stored email and other electronic communication); US PATRIOT Act 2001.

Privacy versus security when monitoring or censoring the Internet

This is the most likely essay on this topic. The slides give you the building blocks: the conundrum (balance), government surveillance, the Patriot Act and its critics, anonymity, and (in the later slides) workplace monitoring and surveillance cameras. The answer below uses them and adds a few points from general knowledge.

> **Q:** QB A2 · 8 marks **What are the main ethical issues in sustaining a delicate balance between Privacy and Security issues when monitoring/censoring use of the internet technologies? Discuss with examples.**

> **How the marks come.** 8 marks means about 4 issues with an example each, plus a short balancing introduction and conclusion. Name the issue first, in the marker words, then the example. Do not write a general essay on privacy without examples: the question says "with examples".

Case: Moxie Marlinspike and the Saudi surveillance request

The case (2013/14 paper, Q3d, 5 marks; the exact question is below). Moxie Marlinspike is a cyber-security expert based in San Francisco. He describes himself as having worked as a software engineer, hacker, sailor, captain and shipwright. The Wall Street Journal identified him as chief technology officer and co-founder of Whisper Systems, which makes privacy and security software.

In May 2013 he wrote on his blog that an employee of a Saudi Arabian telecommunications company had contacted him by email. The employee wanted his help setting up a surveillance programme at the request of the Saudi government. The programme was meant to monitor communications on Twitter, WhatsApp, Viber and Line (apps that allow users to make calls and send texts). The employee sent design documents suggesting tactics such as compelling a certificate authority in the jurisdiction of the UAE or Saudi Arabia to produce SSL certificates they could use for interception, and purchasing SSL vulnerabilities or other exploits.

Marlinspike asked questions to understand the programme better, then declined to help. The contact replied that Saudi Arabia was trying to respond to an ongoing terrorist threat and that this was why he sought help. He added that if Marlinspike was not interested, perhaps he was indirectly helping those who curb freedom with their brutal activities. Marlinspike wrote that this kind of surveillance is "currently happening everywhere". The case text is by Irina Raicu.

The certificate authority link: in public-key systems a Certificate Authority vouches for a site's identity (see encryption). If an authority issues false certificates, an interceptor can pose as the site and read supposedly secure traffic.

> **Q:** 2013/14 paper Q3d · 5 marks **Assuming that all the details of Marlinspike's account are correct, did Marlinspike act ethically in rejecting the request from the telecom? Why, or why not?**

Case: smart lampposts and surveillance in cities (CAT 1 2021-22, Q3)

The case. Smart lampposts carry Internet-of-Things (IoT) sensors. In Hong Kong (article of August 2018) they collect real-time city data: weather, air quality, temperature, people and vehicle flow. They also provide wifi hotspots, electric vehicle charging, information dashboards, real-time traffic updates and parking-space information. They are adopted to cut energy use and raise public safety. Lights dim when streets are empty and brighten when motion sensors trigger.

- **United States.** Los Angeles is equipping streetlights with sensors that can detect gunshots and other noises that may threaten public safety. Schenectady, New York, installed LED lights in 5,000 streetlights, managed through a secure web browser. Its mayor said that if you walk your dog at 10 pm and the lights come on, you get an added level of security. San Jose, California, approved 1,000 smart streetlights in 2018 that detect car crashes, gunshots and earthquakes. An NBC report said they also have facial recognition, cameras and audio recording. A councilman raised privacy concerns, while one resident said that if people know they are being watched, they do not do stupid things.
- **Singapore.** The Smart Nation Sensor Platform (2018) plans to fit all 100,000 lampposts with sensors. Privacy concerns were raised about facial recognition combined with high-definition cameras. An official said there were no plans to probe into how people live or to do "moral policing". A local professor said cyber security and data leaks could be the bigger dangers.
- **Hong Kong rollout.** The rollout began in July 2019. During mass protests that August, demonstrators worried about government surveillance cut down the new lampposts. The government admitted the hardware could spy on citizens but said the fears were unfounded. Officials promised to disable some features (licence-plate recognition, continuous audiovisual surveillance) and said the posts could not do the invasive facial recognition used in China. After 20 lampposts were destroyed, a local company ended its contract after its staff and executives' families received threats. It hoped to "reassure the public" and noted that the public had grown cautious and mistrustful of technology.

The question has six parts (a to f). The model answer below is built from the lecture's surveillance material (conundrum, anonymity, abuse of power, checks and balances, advanced surveillance technology) and the solution file's structure.

Q: `CAT 1 2021-22 Q3a` **Who are the stakeholders involved in the development, deployment, and use of smart lampposts?**

Q: `CAT 1 2021-22 Q3b` **Who should be consulted about such projects' goals and development?**

Q: `CAT 1 2021-22 Q3c` **Identify two dilemmas that you recognize from the above case study.**

Q: `CAT 1 2021-22 Q3d` **What steps might business and government teams need to take to access the information/perspectives needed to manage the ethical landscape of such projects?**

Q: `CAT 1 2021-22 Q3e` **What are some ethical issues that should be addressed as part of the deployment of smart lampposts in cities? Do those issues differ from city to city?**

Q: `CAT 1 2021-22 Q3f` **How might the deployment of smart lampposts be evaluated through the ethical lenses of rights, justice/fairness, utilitarianism, common good, and virtue ethics?**

Check against the slides. The solution file's answers use the general vocabulary above (stakeholders, privacy impact assessment, ethical lenses). None of it contradicts the lecture. The lecture itself only supplies the privacy conundrum, the surveillance advocates-versus-critics argument and the facial recognition example. Treat the stakeholder and consultation lists as exam-style model content, not slide content.

MARKER WORDS: PRIVACY VERSUS SECURITY AND SURVEILLANCE CASES

Say it like this:

> Surveillance offers **security** but threatens the **right to privacy**, so the ethical task is a **balance**: **reasonable limits**, **transparency** and **oversight** to prevent **abuse** of surveillance powers.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
balance	opening and closing any privacy-versus-security answer	**MUST**
anonymity	explaining what monitoring and decryption remove	**MUST**
abuse	explaining the danger of unchecked power	**MUST**
civil liberties	explaining harm to speech, assembly and movement	**MUST**
transparency	recommending how to run surveillance ethically	**MUST**
oversight	recommending independent checks	**MUST**
intimidation of political dissenters	giving the main risk of surveillance technology	**SHOULD**
facial recognition	giving a modern surveillance example	**MUST**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The five ethical lenses in the smart-lamppost question (from the question, not the slides).
Rights; justice/fairness; utilitarianism; common good; virtue ethics.

Surveillance advocates versus critics (advanced surveillance slide).
Advocates: people have no legitimate expectation of privacy in public places. Critics: potential for abuse, such as intimidation of political dissenters and blackmail of people caught with the "wrong" person or in the "wrong" place.

Four facts about Patriot Act powers and limits (slide facts).
Sweeping powers; wiretaps without court order; removes checks and balances so courts cannot stop abuse; sunset provisions (terminated 12/31/05).

Which two laws did the sunset provisions primarily modify?
Title III and FISA (they allow the government to conduct wiretaps without a court order).

Encryption and identity theft

~25 min to read | Low priority | 4 questions

Source: Set 7, slides 9-16. Earlier privacy material is in section t18; profiling, workplace monitoring and spam are in section t20.

> **Priority.** No past paper question has been found on these slides. The practice questions below are original, written in the question bank style. Learn the definitions word for word; they are short.

Cryptography and encryption

Cryptography is the science of encoding messages so that only the sender and the intended receiver can understand them. It is the key tool for ensuring three things about electronic messages and online business transactions: **confidentiality** (nobody else can read the message), **integrity** (the message was not changed on the way) and **authenticity** (the message really comes from who it claims to come from).

Encryption is the process of converting electronic messages into a form understood only by the intended recipients. Plain text goes in, unreadable cipher text comes out, and only the holder of the right key can turn it back.

> **Key point.** Cryptography is the science. Encryption is the process. Do not swap the two in a definition.

MARKER WORDS: CRYPTOGRAPHY AND ENCRYPTION

Say it like this:

> Cryptography is the **science of encoding messages** so that **only the sender and intended receiver** can understand them. Encryption is the **process of converting electronic messages** into a form understood **only by the intended recipients**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
confidentiality	saying what cryptography protects	**MUST**
integrity	saying a message was not altered	**SHOULD**
authenticity	saying the sender is who they claim to be	**SHOULD**
online business transactions	giving the use of cryptography	**SHOULD**
intended recipients	defining encryption	**SHOULD**

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Define cryptography.
The science of encoding messages so that only the sender and intended receiver can understand the messages.

The three things cryptography ensures for electronic messages and online transactions.
Confidentiality, integrity, authenticity.

Define encryption.
The process of converting electronic messages into a form understood only by the intended recipients.

Encryption keys, public-key and private-key systems

An **encryption key** is a (large random) value applied using an algorithm to encrypt or decrypt text. The **length of the key determines the strength** of the encryption algorithm: a longer key has more possible values, so it is harder to guess.

Public-key encryption system

This system uses **two keys**, a public key and a private key, both belonging to the message recipient.

- **The recipient's public key** is readily available. The sender uses it to encrypt the message.
- **The recipient's private key** is mathematically related to the public key. It is kept secret and used for decryption.

RSA is a public-key encryption algorithm. RSA keys are typically 1024 to 2048 bits long.

The public-key diagram (slide 11, a picture)

The picture shows the flow of a message under public-key encryption:

1. The sender must **obtain the recipient's public key from a Certificate Authority**. The Certificate Authority is a trusted third party that vouches that the public key really belongs to that recipient.
2. The sender uses that public key to encrypt the message. It travels across the network as unreadable cipher text.
3. The recipient decrypts it with their own private key. **Only the recipient can read the message**, because only they hold the private key.

beyond the slides The Certificate Authority issues a digital certificate that ties a public key to a named person or organisation. Without it, an attacker could hand the sender a fake public key and read the message.

Private-key encryption system

A private-key (symmetric) system uses a **single key to encode and decode** messages. Its big problem is **distributing the key secretly** to the sender and the receiver. If the key is intercepted on the way, the security is gone. This is the issue the slides call paramount, and it is the reason public-key systems were invented.

	Public-key system	**Private-key system**
Number of keys	Two: public and private	One single key
Encrypting	Recipient's public key, openly available	The shared key
Decrypting	Recipient's private key, kept secret	The same shared key
Main issue	Sender must obtain a genuine public key (Certificate Authority)	Secretly distributing the key to sender and receiver
Example	RSA	Not named in the slides

Q: Practice · 4 marks **Distinguish between public-key and private-key encryption.**

MARKER WORDS: KEYS, PUBLIC-KEY AND PRIVATE-KEY SYSTEMS

Say it like this:

> A public-key system uses **two keys**: the recipient's **public key**, readily available and used for **encryption**, and the recipient's **private key**, kept secret and used for **decryption**. A private-key system uses a **single key**, so **secretly distributing the key** is the main issue.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
length of key determines strength	explaining why long keys are used	**SHOULD**
algorithm	defining an encryption key	**SHOULD**
mathematically related	describing the public and private key pair	**SHOULD**
Certificate Authority	saying where the public key comes from	**SHOULD**
RSA	giving an example of a public-key algorithm	**SHOULD**
distributing the key secretly	giving the weakness of private-key systems	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

What is an encryption key?
A (large random) value applied using an algorithm to encrypt or decrypt text.

What determines the strength of an encryption algorithm?
The length of the key.

Describe a public-key encryption system.
It uses two keys: the recipient's public key (readily available, used for encryption) and the recipient's private key (mathematically related to the public key, kept secret, used for decryption).

What is RSA?
A public-key encryption algorithm. RSA keys are typically 1024-2048 bits long.

Describe a private-key encryption system and its main issue.
A single key to encode and decode messages; the issue of secretly distributing the private key to sender and receiver is paramount.

Building encryption into systems, and export control

Encryption brings **management and administration headaches**: keys must be generated, stored, shared and recovered if lost. Despite this, **most people agree encryption must eventually be built into** three places:

- networks;
- file servers;
- tape backup systems.

The slides give one product example. A **Seagate Technology hard drive automatically encrypts all data**, and the user **must know the password** to access the data. If the drive is lost or stolen, the data is useless to the thief.

Governments also control encryption. The **U.S. Arms Export Control Act** controls the export of encryption technology, hardware and software. **Violators face a 10-year jail term and a $1M fine.**

beyond the slides Ethical tension for discussion: strong encryption protects privacy and business secrets, but it also lets criminals hide communications from law enforcement. This is why governments restrict exports and sometimes demand access.

Q: Practice · 6 marks **Discuss the ethical issues raised by the use of strong encryption, considering both privacy and law enforcement.**

MARKER WORDS: ENCRYPTION IN PRACTICE

Say it like this:

> Encryption **must eventually be built into networks, file servers and tape backup systems**, despite **management and administration headaches**. The **Arms Export Control Act** controls the export of encryption technology, hardware and software.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
networks, file servers, tape backup systems	saying where encryption should be built in	SHOULD
management and administration headaches	giving a cost of encryption	SHOULD
automatically encrypts all data	describing the Seagate drive	SHOULD
must know password	explaining how the drive protects data	SHOULD
Arms Export Control Act	giving the law on exporting encryption	SHOULD
10-year jail term and $1M fine	giving the penalty for violators	SHOULD

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Three places encryption must eventually be built into.
Networks; file servers; tape backup systems.

What does the Seagate Technology hard drive do?
Automatically encrypts all data; the user must know the password to access data.

What does the U.S. Arms Export Control Act control, and what is the penalty?
It controls the export of encryption technology, hardware, and software. Violators face a 10-year jail term and a $1M fine.

Identity theft

Identity theft is the theft of key pieces of personal information to gain access to a person's financial accounts. With this information the thief may apply for new credit or financial accounts, register for college courses and so on, all in someone else's name.

The information stolen includes:

- name;
- address;
- date of birth;
- Social Security number;
- passport number;
- driver's license number;
- mother's maiden name.

Identity theft is the **fastest growing form of fraud in the United States**. Victims spend **more than 600 hours over several years** recovering from it. The slides also criticise the **lack of initiative by companies in informing people** whose data was stolen.

Example: Stony Brook University. The personal information of 90,000 people in a Stony Brook University database was accidentally posted to Google and left there until it was discovered almost two weeks later. No hacker was needed: carelessness by the organisation exposed the data.

Phishing and spear-phishing

Phishing is an attempt to steal personal identity data by tricking users into entering information on a counterfeit web site. The lure is usually a spoof e-mail.

Spear-phishing is a variation in which employees are sent phony e-mails that look like they came from high-level executives within their own organisation. It differs from ordinary phishing because it is targeted at specific people in one organisation and uses a trusted internal name, so it is more convincing.

The phishing e-mail (slide 16, a picture). The slide shows a sample e-mail used by phishers: a message that looks like it comes from a bank or online service and asks the reader to follow a link and enter account details.

beyond the slides Typical tell-tale signs of a phishing e-mail: a generic greeting such as "Dear customer"; an urgent threat, for example "your account will be closed in 24 hours"; a sender address that does not match the real organisation's domain; a link whose real address differs from the text shown; spelling and grammar errors; a request for a password, PIN or Social Security number, which genuine organisations do not ask for by e-mail; and an unexpected attachment.

Spyware and the law

Spyware is keystroke-logging software downloaded to a user's computer without consent. It captures account usernames, passwords, credit card numbers and other sensitive information. It **operates even if the**

infected computer is not connected to the Internet: it records keystrokes until the user reconnects, and the data collected is then e-mailed to the spy or posted to a web site.

The **Identity Theft and Assumption Deterrence Act of 1998** was passed to fight identity fraud. It makes identity theft a **Federal felony, punishable by 3 to 25 years in prison**.

Q: **Practice · 4 marks** How does spear-phishing differ from phishing? Name one other way identity theft is carried out.

Q: **Practice · 5 marks** Explain what is meant by identity theft. State the information that is stolen and describe two ways in which the problem can be reduced.

MARKER WORDS: IDENTITY THEFT

Say it like this:

> Identity theft is the **theft of key pieces of personal information** to gain access to a person's **financial accounts**. Phishing is an attempt to steal personal identity data by **tricking users** into entering information on a **counterfeit Web site**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
identity theft	naming the crime	MUST
personal information	saying what is stolen	MUST
fastest growing form of fraud	stressing the scale of the problem	SHOULD
more than 600 hours	describing the cost to victims	SHOULD
counterfeit Web site / spoof e-mails	defining phishing	SHOULD
high-level executives	defining spear-phishing	SHOULD
keystroke-logging	defining spyware	SHOULD
Federal felony (3-25 years)	giving the legal penalty	SHOULD

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Define identity theft.
Theft of key pieces of personal information to gain access to a person's financial accounts.

The personal information that identity thieves steal (7 items).
Name; address; date of birth; Social Security number; passport number; driver's license number; mother's maiden name.

Define phishing.
An attempt to steal personal identity data by tricking users into entering information on a counterfeit Web site (spoof emails).

Define spear-phishing.
A variation of phishing in which employees are sent phony e-mails that look like they came from high-level executives within their organization.

Define spyware (identity theft context).
Keystroke-logging software downloaded to a user's computer without consent; it captures usernames, passwords, credit card numbers and other sensitive information, even when the computer is not connected to the Internet.

Identity Theft and Assumption Deterrence Act of 1998.
Passed to fight identity fraud; makes it a Federal felony (3-25 years in prison).

Profiling, workplace monitoring, spam and surveillance technology

~40 min to read | Medium priority | 5 questions

Source: Set 7, slides 17-29. Earlier privacy material (privacy law, government surveillance) is in section t18; encryption and identity theft are in section t19. The social-media case of Simon (QB A1(i)) is answered with the 7-step template in section t07.

> **How this section fits together.** Slides 17-20 explain how companies profile consumers. Slides 21-22 say how to treat that data responsibly. Slide 23 covers employers watching employees. Slides 24-25 cover spam. Slides 26-27 cover cameras, facial recognition and GPS. Slides 28-29 are summaries. Every one of these topics has the same shape: a useful capability set against a privacy cost. That is the "ethical conundrum" the lecturer keeps naming, and it is the frame for every essay answer here.

Consumer profiling: how data is collected (slides 17-19)

The opening claim of the slides is that **companies can collect information about consumers without their explicit permission**. Some collection is open. Companies openly collect personal information about Internet users when they register at web sites, complete surveys, fill out forms or enter contests online. The user knows they typed something in. The hidden collection is the problem.

Cookies and tracking software

- **Cookies** are text files a web site places on the user's hard drive so that it can remember information. Examples: site preferences and the contents of an electronic shopping cart.
- The browser sends the cookie back to the server **unchanged** each time it accesses that server. This is how the site recognises you on your next visit.
- **Tracking software** identifies visitors to your web site, for example from pay-per-click accounts. The site owner learns where the visitor came from.

Offline profiling

Similar methods are used outside the Web. Marketing firms **warehouse consumer data**: credit card purchases, frequent flier points, mail-order catalogue purchases and phone surveys. The result is databases that contain a huge amount of consumer behavioural data. So profiling is not only a web problem. Online and offline records can be combined.

Affiliated web sites and DoubleClick

An **affiliated web site** group is a group of web sites served by a single advertising network. **DoubleClick** was the slides' example. It tracks ad clicks and web purchases across the group, which is useful for marketers and sellers. Because one network sees your visits to many sites, it can build a much fuller picture of you than any single site could. Marketers use cookies to recognise return visitors and store useful information about them, which gives **customised service for each consumer**. The same mechanism that makes service convenient is the one that builds the profile.

Types of data collected while surfing

Data type	What it is
GET data	The affiliated web sites visited and the information requested.
POST data	Form data: what you typed into forms.
Click-stream data	Monitoring of consumer surfing activity: the trail of clicks.

Four ways to limit or stop cookies

1. Set the browser to limit or stop cookies.
2. Manually delete them from the hard drive.
3. Download and install a cookie-management program.

4. Use anonymous browsing programs that do not accept cookies. Example: anonymizer.com lets you hide your identity while browsing.

FLASHCARDS: CLICK TO FLIP

What are cookies?
Text files a web site places on the user's hard drive so that it can remember information (for example site preferences, contents of a shopping cart). The browser sends them back to the server unchanged each time it accesses that server.

What is an affiliated web site group? Give the slides' example.
A group of web sites served by a single advertising network. Example: DoubleClick, which tracks ad clicks and web purchases.

The three types of data collected while surfing the Web.
GET data (affiliated web sites visited and information requested); POST data (form data); click-stream data (monitoring of consumer surfing activity).

Four ways to limit or stop the deposit of cookies.
Set the browser to limit or stop cookies; manually delete them from the hard drive; install a cookie-management program; use anonymous browsing programs that do not accept cookies.

Personalization software and P3P (slide 20)

Marketers use **personalization software** to optimise the number, frequency and mixture of their ad placements. There are four kinds.

Type	How it decides what to show you
Rules-based	Uses business rules tied to customer-provided preferences or online behaviour to determine the most appropriate page views.
Collaborative filtering	Gives consumer recommendations based on products purchased by customers with similar buying habits.
Demographic filtering	Considers user zip codes, age and sex when making product suggestions.
Contextual commerce	Associates product promotions or ads with the content the user is currently viewing.

Quick way to tell them apart: rules = "if the customer did X, show Y"; collaborative = "people like you bought"; demographic = "people of your age, sex and area"; contextual = "ads that match the page you are on".

Platform for Privacy Preferences (P3P)

P3P is the technical defence. It **shields users from sites that do not provide the desired level of privacy protection**. P3P software in a browser downloads the privacy policy for each site visited and **notifies the user if the policy does not match their preferences**. The user sets preferences once; the browser compares them with each site's policy.

FLASHCARDS: CLICK TO FLIP

The four types of personalization software.
Rules-based; collaborative filtering; demographic filtering; contextual commerce.

Collaborative filtering.
Consumer recommendations based on products purchased by customers with similar buying habits.

Contextual commerce.
Associates product promotions/ads with content the user is currently viewing.

What does P3P do?
Shields users from sites that do not provide the desired level of privacy protection. P3P software in a browser downloads the privacy policy of each site visited and notifies users if the policy does not match their preferences.

MARKER WORDS: CONSUMER PROFILING

Say it like this:

> Consumer profiling is the collection of information about consumers, often **without their explicit permission**, using **cookies**, **tracking software** and **click-stream data**, to build a profile for **customised service** and targeted advertising.

Vocabulary to weave into explanations:

Marker	Use it when you are…	Tag
without their explicit permission	saying why profiling is a privacy problem	**SHOULD**
cookies (text files on the hard drive)	explaining how a site recognises return visitors	**SHOULD**
click-stream data, GET data, POST data	listing what is collected while surfing	**SHOULD**
affiliated web sites / advertising network	explaining how one firm sees many sites	**SHOULD**
data mining	linking profiling to the exam question on data mining	**MUST**
informed consent	saying what users are denied by hidden collection	**MUST**
P3P / privacy preferences	giving a technical safeguard	**SHOULD**

▶ **Recall drill: write the definition from memory, then click each blank to check**

Data mining and privacy

beyond the slides The slides do not define data mining, but profiling is its everyday form: large stores of consumer data (the warehoused credit card, mail-order and click-stream records above) are searched by software for patterns about individuals and groups. The privacy issues follow from that.

Q: **QB D3 · 5 marks (also sol_exam2122 Q4c)** **What ethical issues does the use of data mining raise with respect to privacy?**

Treating consumer data responsibly (slides 21-22)

The slide's opening: **strong measures are required to avoid customer relationship problems**. Customers who feel watched leave, complain or sue. The recognised standard is the **Code of Fair Information Practices and the 1980 OECD privacy guidelines**. Under them a company should:

1. **Collect only the personal information necessary** to deliver its products or services.
2. **Protect** this information.
3. **Inform customers** if it intends to use the information for research or marketing.
4. Provide a means for customers to **opt out**.

The summary slide (29) shortens this to: to protect consumer data, organisations only collect personal information necessary to deliver the product or service.

Chief privacy officer (CPO)

A CPO is an **executive who oversees data privacy policies and initiatives**. The role **avoids violating government regulations** and **assures customers that their privacy will be protected**.

The manager's checklist (slide 22)

Slide 22, "Manager's Checklist for Treating Consumer Data Responsibly", is a table of eight questions, each answered Yes or No. It is the practical form of slide 21. A responsible manager should be able to answer Yes to all eight:

1. Do you have a written data privacy policy that is followed?
2. Can consumers easily view your data privacy policy?
3. Are consumers given an opportunity to opt in or opt out of your data policy?
4. Do you collect only the personal information needed to deliver your product or service?
5. Do you ensure that the information is carefully protected and accessible only by those with a need to know?
6. Do you provide a process for consumers to review their own data and make corrections?
7. Do you inform your customers if you intend to use their information for research or marketing and provide a means for them to opt out?
8. Have you identified a person who has full responsibility for implementing your data policy and dealing with consumer data issues?

Notice what the checklist adds to slide 21: a **written, followed, easily viewable** privacy policy, a chance to **opt in** as well as opt out, a way for consumers to **review and correct** their own data, and **access only by those with a need to know**.

FLASHCARDS: CLICK TO FLIP

Manager's checklist for treating consumer data responsibly: the eight questions (slide 22).
Written data privacy policy that is followed; consumers can easily view it; opportunity to opt in or opt out; collect only the information needed; information protected and accessible only to those with a need to know; process for consumers to review their data and make corrections; inform customers of research or marketing use with a means to opt out; a named person with full responsibility for the data policy.

What a company should do under the Code of Fair Information Practices and 1980 OECD guidelines (4 points).
Collect only the personal information necessary to deliver its products/services; protect this information; inform customers if it intends to use it for research or marketing; provide a means for customers to opt out.

Chief privacy officer (CPO).
An executive to oversee data privacy policies and initiatives; avoids violating government regulations and assures customers that their privacy will be protected.

MARKER WORDS: TREATING CONSUMER DATA RESPONSIBLY

Say it like this:

> Under the **Code of Fair Information Practices** and the **1980 OECD privacy guidelines**, a company should **collect only personal information necessary** to deliver its product or service, **protect** it, **inform customers** of research or marketing use, and let them **opt out**. A **chief privacy officer** oversees this.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
Code of Fair Information Practices	naming the standard for handling consumer data	SHOULD
1980 OECD privacy guidelines	giving the source of the standard	SHOULD
only personal information necessary	stating the data-minimisation rule	SHOULD
opt out	giving customers control	SHOULD
chief privacy officer (CPO)	saying who is accountable in the firm	SHOULD
consent	explaining why customers must be informed	MUST

▶ **Recall drill: write the sentence from memory, then click each blank to check**

Workplace monitoring (slide 23)

The **ethical conundrum**: how do you ensure worker productivity without violating the privacy rights of employees? Employers monitor workers by recording email, surfing activity and files, and even by videotaping employees on the job. The employer's aim is to ensure that the corporate IT usage policy is followed.

The legal position (US, as in the slides). The Fourth Amendment cannot be used to limit how a **private** employer treats its employees. **Public-sector** employees have far greater privacy rights, because of the **"reasonable expectation of privacy"** test from **Katz v. U.S.** (the slide gives it as a Supreme Court ruling, dated 1998 on the slide). A private employee therefore has little constitutional protection against the employer.

> **Watch out.** The slide writes "1998" for Katz v. US. Reproduce the slide's wording in the exam. beyond the slides The historical Katz decision is from 1967, which is where the "reasonable expectation of privacy" idea comes from. If you mention a date, hedge or leave it out.

What privacy advocates want. Federal legislation to keep employers from infringing employees' privacy rights; employees to be informed of electronic monitoring devices; and a restriction on the type of information collected.

beyond the slides In Uganda the relevant protection is Article 27 of the Constitution (right to privacy of person, home and other property and communications) and the Data Protection and Privacy Act, 2019, which requires data to be collected for a lawful purpose, with consent or another legal basis, and limits what is kept. Name them only as supporting points.

Q: QB A1(ii) · 5 marks **Simon is a social media consultant. The CIO asks him to collect personal data on the staff in the IT department via Facebook and LinkedIn. How ethical or legal is it for the CIO to use employees' personal social media activity in the HR monitoring process?**

Q: 2014 and 2015 papers Q5(c) · 4 marks **Identify two ways in which a company may violate an employee's right to privacy.**

Herman case (2014/15 Q5d). That question (8 marks) reads:

> EXAM WORDING
> With reference to Herman's case, comment on the view that employers do not have responsibilities (to employees) beyond those of the employment contract.

In the case, Herman works more than 10 hours a day and six days a week, 300 km from his family, and his contract states no working hours or days. It is about employer duties and work-life balance, not privacy. It is not covered here; see t09 and t10 for employer duties and contracts. Q5(b) (6 marks:

> EXAM WORDING
> Identify and explain three (3) ethical responsibilities of employers towards employees

), sits with the same material.

FLASHCARDS: CLICK TO FLIP

The ethical conundrum of workplace monitoring.
Ensuring worker productivity without violating the privacy rights of employees.

Why can the Fourth Amendment not protect most employees from monitoring?
It cannot be used to limit how a private employer treats its employees. Public-sector employees have far greater privacy rights: the "reasonable expectation of privacy" (Katz v. U.S.).

What privacy advocates want on workplace monitoring.
Federal legislation to keep employers from infringing employees' privacy rights; inform employees of electronic monitoring devices; restrict the type of information collected.

MARKER WORDS: WORKPLACE MONITORING

Say it like this:

> Workplace monitoring raises the **ethical conundrum** of ensuring **worker productivity** without violating the **privacy rights of employees**. Employers record **email, surfing activity and files** to ensure the **corporate IT usage policy** is followed, and employees should be **informed**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
ethical conundrum	opening any monitoring answer	**SHOULD**
productivity	giving the employer's side	**MUST**
corporate IT usage policy	saying why employers monitor	**SHOULD**
reasonable expectation of privacy	giving the legal test (Katz v. U.S.)	**SHOULD**
private employer vs public-sector employees	explaining the legal gap	**SHOULD**
inform employees of monitoring	recommending what employers should do	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

Spamming (slides 24-25)

Spamming is the transmission of the same e-mail message to a large number of people. It is an extremely inexpensive method of marketing. Compared with a direct-mail campaign:

	Spam campaign	Direct-mail campaign
Cost	$1K	$10K
Time to develop	3 weeks	3 months
Time for feedback	48 hours	3 weeks

Many **legitimate organisations** use it, for example for product announcements. But it can also contain **unwanted and objectionable material**. The slide says these last two points are the ethical conundrum: a cheap, legitimate tool that also floods people with unwanted content.

Statistics on the slide:

- Spam is 40% of all email.
- 12.4 billion spam emails are sent daily.
- Each person receives 6 spam emails a day, about 2,200 a year.
- Cost to all non-corporate Internet users: $255 million.
- Cost to all U.S. corporations in 2002: $8.9 billion.
- 26 states had anti-spam laws.

The CAN-SPAM Act 2004

Full name: **Controlling the Assault of Non-Solicited Pornography and Marketing Act**. It says it is **legal to spam but**:

- spammers **cannot disguise their identity**;
- there must be a **label** in the message specifying that the e-mail is an ad or solicitation;
- they must include a way for recipients to indicate they do not want future mass mailings (**opt out**).

Criticism: the Act **may have actually increased the flow of spam**, because it legalises the sending of unsolicited e-mail.

FLASHCARDS: CLICK TO FLIP

What is spamming?
Transmission of the same e-mail message to a large number of people.

What the CAN-SPAM Act 2004 requires of spammers (it says spamming is legal but...).
They cannot disguise their identity; there must be a label specifying the e-mail is an ad or solicitation; they must include a way for recipients to opt out of future mass mailings.

The main criticism of CAN-SPAM.
It may have actually increased the flow of spam, as it legalises the sending of unsolicited e-mail.

Spam versus direct mail: cost, development time, feedback time.
$1K vs $10K; 3 weeks vs 3 months; 48 hours vs 3 weeks.

MARKER WORDS: SPAMMING

Say it like this:

> **Spamming** is the **transmission of the same e-mail message to a large number of people**. It is an **extremely inexpensive method of marketing** but can contain **unwanted and objectionable materials**. The **CAN-SPAM Act 2004** makes spam legal if the sender does not **disguise identity**, labels the message as an ad and offers an **opt out**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
inexpensive method of marketing	giving the sender's benefit	**SHOULD**
legitimate organisations / product announcements	showing spam is not always criminal	**SHOULD**
unwanted and objectionable material	giving the harm	**SHOULD**
cannot disguise their identity	listing CAN-SPAM conditions	**SHOULD**
label (ad or solicitation)	listing CAN-SPAM conditions	**SHOULD**
opt out	listing CAN-SPAM conditions	**SHOULD**
legalises unsolicited e-mail	criticising the Act	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

Advanced surveillance technology (slides 26-27)

These tools provide **exciting new data-gathering capabilities versus personal-privacy issues**. The two sides:

- **Advocates:** people have no legitimate expectation of privacy in public places.
- **Critics:** it creates potential for abuse, such as intimidation of political dissenters and blackmail of people caught with the "wrong" person or in the "wrong" place.

Camera surveillance

U.S. cities plan to expand surveillance systems. London has one of the world's largest public surveillance systems. A **"smart surveillance system"** singles out people acting suspiciously, so a computer, not a human, decides who is watched. (The smart-city lamppost case is in t18.)

Facial recognition software

It identifies criminal suspects and other undesirable characters, but it **yields mixed results**. At **Boston's Logan airport** it produced **96 failures and 153 successes**. That is 249 trials, so roughly 61% worked and about 39% failed. (Do not claim more than the slide: it shows the technology works often but not reliably.) A failure means either a suspect is missed or an innocent person is flagged.

GPS chips

Global Positioning System chips are placed in many devices, such as cars and cellphones, to precisely locate users.

- **Good:** accurately respond to 911 callers; real-time, location-aware marketing.
- **Bad:** wireless spamming from local restaurants and similar businesses; your whereabouts are always known.

Q: **QB D4 · 4 marks (also CAT 2021-22 Q4d)** **What ethical issues does the use of facial recognition systems raise with respect to privacy?**

Q: **QB D5 · 8 marks (also CAT 2021-22 Q4e, which asks for two reasons)** **"Honest, upright, law-abiding citizens have nothing to fear from the distribution of their personal data." Do you agree or not? Give at least three reasons for your position and one reasonable objection to it.**

FLASHCARDS: CLICK TO FLIP

The two sides of the surveillance debate (slide wording).
Advocates: people have no legitimate expectation of privacy in public places. Critics: creates potential for abuse - intimidation of political dissenters, blackmail of people caught with the "wrong" person or in the "wrong" place.

What is a "smart surveillance system"?
A camera surveillance system that singles out people acting suspiciously.

The Logan airport facial recognition result.
Mixed results at Boston's Logan airport: 96 failures, 153 successes.

The good and the bad of GPS chips (slide wording).
Good: accurately respond to 911 callers; real-time location-aware marketing. Bad: wireless spamming from local restaurants etc.; your whereabouts are always known.

MARKER WORDS: ADVANCED SURVEILLANCE TECHNOLOGY

Say it like this:

> Advanced surveillance technology gives **exciting new data-gathering capabilities** but raises **personal-privacy issues**: advocates say there is **no legitimate expectation of privacy in public places**, while critics warn of **potential for abuse**, such as **intimidation of political dissenters** and **blackmail**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
facial recognition	answering D4, D5 or any surveillance question	**MUST**
informed consent	saying people are scanned or mined without agreeing	**MUST**
bias / discrimination	criticising accuracy and fairness	**MUST**
smart surveillance system	describing computer-flagged suspicion	**SHOULD**
96 failures, 153 successes	giving evidence that recognition is unreliable	**SHOULD**
potential for abuse	stating the critics' case	**SHOULD**
GPS: "whereabouts always known"	giving a location-tracking example	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

Summary of the privacy topic (slides 28-29)

Slide 28 recaps earlier material: the legal right to privacy has four aspects (web-surfing privacy, identity theft, HIPAA, false information about someone on the web). Laws of the past 40 years affect privacy: FISA 1978 (wiretapping aliens), ECPA 1986 (standards for access to stored email and other electronic communication) and the US PATRIOT Act 2001. These and other laws authorise government electronic surveillance. Data encryption has public key and private key systems, and identity theft is the last item. See t18 and t19.

Slide 29 recaps this section: consumer behaviour data is collected both online and offline (consumer profiling); the Code of Fair Information Practices and 1980 OECD guidelines say organisations should only collect personal information necessary to deliver the product or service; employers record and review employee communications and activities on the job (workplace monitoring); and information technology advances for surveillance include surveillance cameras, facial recognition software and GPS systems.

FLASHCARDS: CLICK TO FLIP

Four aspects of the legal right to privacy (slide 28; see t18).
Web-surfing privacy; identity theft; HIPAA; false information about someone on the web.

Key point. In an essay on any of these topics, use the same four moves: state the benefit, state the privacy cost, name the standard that balances them (fair information practices, informed consent, notice and opt-out) and conclude with a recommendation.